

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Designed and trained **Deep Learning networks** for object classification tasks, achieving a **92% accuracy** on a benchmark dataset.
- Developed Python scripts to automate data acquisition and preprocessing, ensuring **data integrity** for model training pipelines and reducing manual effort by **40%**.
- Performed statistical analysis on model performance metrics and fine-tuned hyperparameters, leading to a **15% improvement** in inference speed for real-time applications.
- Implemented **REST APIs** using Java Spring Boot to integrate ML model outputs with downstream enterprise systems, facilitating seamless data flow for operational dashboards.
- Managed production environments on **AWS**, utilizing EC2 and S3 for model deployment and data storage, while ensuring **99.9% uptime** through proactive monitoring with CloudWatch.
- Collaborated with cross-functional teams to define data requirements and validate data pipelines, ensuring alignment with project objectives and stakeholder expectations.

SKILLS

Deep Learning – Keras, PyTorch, scikit-learn, TensorFlow

Computer Vision – OpenCV, 3D object detection, classification, trajectory mapping, sensor fusion

Programming Languages – Python, Java, R

Data & Analytics – Data acquisition, data integrity, data pipelines, statistical analysis, data modelling, Power BI, DAX, Power Query, Tableau

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Linux, Git, Jenkins

Software Engineering – Spring Boot, REST APIs, Microservices, Agile/Scrum

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Automated Lane Detection System

- Developed a Python-based system for real-time **lane detection** using OpenCV and image processing techniques, achieving **95% accuracy** in varied lighting conditions.
- Trained a **Convolutional Neural Network (CNN)** model using PyTorch on a custom dataset of road images to classify lane markings and road edges.

- Integrated the detection module with a simulated vehicle environment, demonstrating its capability for **automated driving systems**.

Vehicle and Pedestrian Classification Model

- Engineered a deep learning model using **Keras** and TensorFlow for accurate classification of vehicles and pedestrians from sensor data.
- Selected and curated diverse datasets, implementing robust data augmentation strategies to improve model generalization and performance across different scenarios.
- Deployed the trained model on an **AWS EC2 instance**, providing an API endpoint for real-time object classification, supporting **1000+ requests per hour**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)